ML Model Accuracy and Performance Report

Dataset Size: 153 Samples Model Type: Hybrid (Isolation Forest + Rule Engine)

1. Confusion Matrix

	Actual Attack	**Actual Safe**
Predicted Attack (Blocked)	**27 (True Positive)**	**5 (False Positive)**
Predicted Safe (Allowed)	**0 (False Negative)**	**121 (True Negative)**

- True Positives (TP): Attacks correctly blocked.
- False Positives (FP): Safe requests incorrectly blocked (due to lowered threshold).
- True Negatives (TN): Safe requests correctly allowed.
- False Negatives (FN): Attacks missed (None observed).

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2. Performance Metrics

A. Accuracy: The percentage of total predictions that were correct.

$$\text{Accuracy} = \frac{TP + TN}{Total} = \frac{27 + 121}{153} = \frac{148}{153}$$

Result: 96.73%

B. Precision: When the model blocks a request, how often is it actually an attack?

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{27}{27 + 5} = \frac{27}{32}$$

Result: 84.38% **(**Note: Precision was impacted by the intentional sensitivity test**.)**

C. **Recall:** Out of all actual attacks, how many did the model catch?

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{27}{27 + 0}$$

Result: 100.00%

D. **F1 Score:** The harmonic mean of Precision and Recall, providing a balanced view.

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Result: 0.915

The model demonstrates perfect Recall (100%), indicating that no malicious payloads bypassed the firewall during testing. While the Precision (84%) reflects the aggressive tuning used during this specific test, the overall Accuracy (96.7%) confirms the system is highly effective for production deployment when tuned to the recommended threshold of 0.50.